OPERATION SHADOW SIGNAL

Student Lab Pack | Module 7: Threat Intelligence and Detection | Set E

Scenario: You are the Security Operations Centre (SOC) team at BlueShield Finance, a fictional Nigerian fintech. A threat bulletin mentions a suspicious IP address, and the account activity logs show several unusual events. Your task is to assess the intelligence, conduct a focused hunt, design and test a detection, and report what the evidence does—and does not—show.

1. Learning objectives

- Separate observed facts from intelligence claims, hypotheses, and unknowns.
- Assess the source reliability and confidence of a threat-intelligence claim.
- Create a focused hunt question and use supplied logs to answer it.
- Map observed behaviour to MITRE ATT&CK only where the evidence supports a mapping.
- Write a simple detection condition and test it with positive and negative cases.
- Explain detection coverage, false positives, false negatives, and rule limitations.
- Produce a concise SOC investigation report with evidence, correction or response recommendations, and verification.

2. Team roles

Role	Responsibility
Threat Intelligence Analyst	Assess the bulletin, source reliability, confidence, and gaps.
Threat Hunter	Write a focused hunt question and analyse the supplied events.
Detection Engineer	Define a detection condition and evaluate its limitations.
Evidence & Reporting Analyst	Maintain an evidence table and prepare the final report.

3. Rules of engagement

- This is a fictional, synthetic-data exercise. Do not scan, contact, or investigate real systems or addresses.
- Treat each supplied event as evidence only for what it explicitly records.
- Do not state that an account is compromised unless the evidence supports that conclusion.
- Record assumptions, missing fields, timestamps, and limitations.
- Use the supplied data only; do not invent events to fill gaps.

4. Threat-intelligence bulletin

Bulletin ID: TI-052

Source description: A bulletin from a verified member of a sector information-sharing group, with a first-seen date and observations from two member organisations.

Claim: The bulletin says that the IP address 198.51.100.150 was used in recent intrusions at financial-sector members, where attackers signed in with valid credentials, added a new MFA method, and created mail-forwarding rules to external addresses.

Evidence included in the bulletin: A campaign description, a first-seen date, and a summary of the post-login behaviour observed by two members are given. No raw logs are supplied, and the bulletin does not say whether this activity has been seen at BlueShield Finance.

Important: All addresses and events in this exercise are reserved/documentation examples or otherwise synthetic. They are not provided as live threat indicators.

5. Synthetic authentication log

Review the following events. The log is deliberately limited: it records sign-in, MFA-method, and mail-rule events only, and does not contain full device, location, user-agent, mailbox-content, or session audit details.

Time	Account	Source IP	Event	Additional note
02:14	ruth	198.51.100.150	Successful login	MFA satisfied by saved-device token
02:19	ruth	198.51.100.150	MFA method added	New authenticator app registered
02:22	ruth	198.51.100.150	Mail rule created	Forward to external address (domain not in company directory)
08:30	ruth	192.0.2.70	Successful login	Office network; MFA push approved
09:00	sola	192.0.2.71	Successful login	Normal office login
09:02	tobi	192.0.2.72	Successful login	Normal office login

6. Activity A — Intelligence assessment (20 minutes)

Work as a team and complete the following:

1. List the claims made by bulletin TI-052 – It was reported that…
    a) The Attacker signed with valid Credentials
    b) Added new MFA\
    c) Created mail-forwarding rules to external addresses.
    d) The IP address 198.51.100.150 was used in the intrusion.
2. Rate the source reliability as High, Medium, or Low and justify your rating.
    Source Reliability Rating – Medium.
    Reason – Claims are from members of the shared organization, and do not supply enough evidence
3. Rate your confidence that the IP is malicious as High, Medium, or Low. Explain why source reliability and confidence in a claim are not the same thing.
    Confidence Rating – High
    Reliability – The trustworthiness of the source, i.e how trustworthy the source is.
    Confidence in claim – How confident the reporter is about his/her claim.
4. Identify at least three pieces of information you would request before treating the bulletin as actionable intelligence.
    A more robust log containing
        - Location of login,
        - user authorization and privileges,
        - Email address for mail rule.
5. Classify each statement below as an observed fact, hypothesis/claim, or unknown:

Statement	Your classification	Evidence or reasoning
ruth's account recorded an MFA method addition and a mail-rule creation from 198.51.100.150 within eight minutes of a successful login from the same address.	Fact	It was reported by a verified member of an organization, and also stated on the log.
The 02:14 login was made by an attacker using stolen credentials.	Unknown	There are no facts to prove a stolen login credential.
ruth knows about, or authorised, the new MFA method and the mail rule.	Hypothesis	There are indications that it came from her account, but there are no clear evidence of her involvement.
The bulletin and the log prove that BlueShield Finance has suffered a confirmed breach and data theft.	Hypothesis	There are indications, but there are no well detailed logs.

7. Activity B — Focused threat hunt (30 minutes)

Write one focused hunt question that can be investigated using the supplied logs. Avoid a question that assumes the account is already compromised.

Our hunt question:

Was Ruth involved in the login, creation of new MFA, and the implementation of a new mail-forwarding rule?_____

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Build a small evidence timeline. Include the event, what it supports, and what it cannot prove.

Time / event	What it supports	What it does not prove / missing evidence
2:14 am/ successful login into Ruth's Account	Login was from ruth's system, and doesn't indicate a password guessing attempt.	• Does not prove that it was an attack. • Does not indicate incident location.
2:19 am/ MFA added	There is an authorization to add a new MFA.	The activity does not prove that Ruth carried out the action.
2:22 am/ Mail rule creation	From the activity, it shows a created mail-forwarding rule to a non-company domain.	The bulletin does not say the activities have been seen or not, we don't know if it's an internal company activity.
8:30 am/ Successful login from Ruth	Login from office network and IP	This activity shows that she was likely not involved in the previous activities.

MITRE ATT&CK mapping: Identify a technique only if the observed behaviour supports it. If the evidence is insufficient, write "insufficient evidence to map confidently" and explain what additional evidence is needed.

Candidate behaviour / technique and rationale:

Candidate behavior – Transfer of mails to a non-company domain.

Tactics – To create a mail rule in favour of his/her objective.

Techniques – Successful login from unknown account and transafer of mails to unknown domain.

Procedures – Successful login, MFA method change, and mail rule creation.

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8. Activity C — Detection engineering and tests (45 minutes)

Design a simple detection rule for repeated failed authentication attempts.

Required condition: Alert if an account has 5 or more failed login attempts within any 10-minute window.

Write the logic in plain English or pseudocode. State which fields are required and what counts as a failed login.

Detection rule:

5>= failed login attempts within 10 minutes would trigger an alert.

Test the rule against each case below. Predict whether it should alert, then record your result and explain why.

Case	Events	Expected alert? (Yes/No)	Reason / observed result
P1	5 failed logins for one account within 3 minutes	Yes	5 failed logins within 10 minutes window (TRUE POSITIVE).
P2	Exactly 5 failed logins for one account within 10 minutes	Yes	5 failed logins within 10 minutes.
N1	A successful login with no failed logins, followed within 10 minutes by a new MFA method and an external mail-forwarding rule	No	A successful login within 10 minutes of attempt (TRUE NEGATIVE).
N2	4 failed logins for one account within 9 minutes	No	Did not complete the 5 attempt rule (FALSE NEGATIVE)... Increase rule time to 10> minutes, reduce attempt trails.
N3	6 failed logins for one account over 30 minutes, with no 10-minute window containing 5 failures	No	Did not complete the 5 attempts rule within 10 minutes (FALSE NEGATIVE).

Explain at least two possible false-positive causes and two ways the detection could miss suspicious activity.

Examples of false positive -

- Five(5) wrong password tried due to possible unnoticed Cap lock.
- An antivirus flagging a self-created malicious test file on a system.

Two ways the detection could miss S.A

- When attacker attempts 4 passwords within 10 minutes.
- When an attacker use stolen ID.

9. Activity D — Incident injects and response (25 minutes)

Your instructor will reveal these updates during the exercise. Discuss how each changes—or does not change—your assessment.

Inject	New information	Your response / what changes?
1	A manager asks: "Does this bulletin plus the log prove we have been breached and customer data stolen?"	It does not show that there is a breach or that customer's credentials have been stolen due to lack of adequate evidence from logs. Therefore nothing changed, it is still just a claim.
2	IT reports that ruth says she was asleep overnight and did not add an MFA method or create a mail rule.	We escalate the incident and do further investigation. What changed? All actions were unauthorized.
3	Your failed-login rule stayed silent during the whole 02:14–02:22 sequence.	Login credentials were accurate. What change? Her details have been leaked and saved in an unknown device.

Prepare a two-minute briefing for management. State the key finding, confidence level, evidence gaps, detection result, and proportionate next steps.

10. Deliverables

- A concise intelligence assessment that distinguishes facts from claims.
- A hunt note with a focused question, timeline, and evidence-based conclusion.
- A detection rule and completed results for all five test cases.
- At least two documented detection limitations and suggested improvements.
- A final SOC investigation report and two-minute verbal briefing.

11. Final SOC report template

Incident / exercise name: Blue Shield Financial Mail Compromise.

Scope and data reviewed: Suspicion breach in company mail rule.

Hunt question: Key Observed fact.

Key observed facts: No Password guessing.

Intelligence assessment (source reliability and confidence): Medium, limited evidence on the logsssprovided.

ATT&CK mapping (or reason mapping is not supported):

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Detection logic and test results: 5 attempts within 10 mins triggers alerts

False-positive / false-negative risks: for false positive => flag and stop a normal event if further investigation is not done

For false negative: low awareness of malicious activities due to lack of alerts

Evidence gaps and limitations: location of the login, Company awareness, foreign domain address

Recommended next steps: isolate ruth from the network, reset password for ruth, investigate the unknow domain.

Conclusion: Ruth account was compromised due to shared or saved login details, further investigation should be carried out on the account so the cases will be escalated to the incidence respond team

12. Assessment rubric (100 points)

Area	Points	What good work demonstrates
Intelligence evaluation	15	Source reliability and confidence are assessed separately; claims are not treated as facts.
Hunt question and evidence	20	Focused question, accurate timeline, and evidence-linked conclusion.
ATT&CK mapping	10	Mapping is justified by observed behaviour, or uncertainty is explicitly stated.
Detection logic and design	20	Rule matches the stated threshold and required fields are identified.
Positive and negative testing	20	All cases are evaluated correctly and results are explained.
Reporting and limitations	15	Clear communication, evidence gaps, limitations, and proportionate next steps.

End of student pack. Submit your team's completed work to your instructor.

MEMBER NAME	POSITION HELD
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